Utica University | Department of Computer Science | Fall 2026

CSC 618: Information Retrieval | Week 10 Assignment

Submission 3: Part C, Questions on the Real-Dataset Experiment

November 2026

I compared TF-IDF, BM25 and a Dirichlet Language Model in Apache Lucene on the Cranfield collection (1400 documents, 225 queries). Methods and code are in Submission 4.

Table 1: Main results (mean over 225 queries, default parameters).

Model	MAP	MRR	P@3	P@10	nDCG@10	Recall (full list)
TF-IDF	0.2931	0.5327	0.3452	0.2244	0.3690	0.9513
BM25	0.2952	0.5204	0.3570	0.2284	0.3741	0.9523
LM-Dirichlet (μ=2000)	0.2341	0.4557	0.2726	0.1840	0.3041	0.9515

1. Did the evaluation code that worked on the controlled dataset also work on the real dataset? What changes were necessary?

Yes. The metric functions in `ir_metrics.py` ran with no changes, and the ranx library gave the same values to six decimals. The changes were all in the code around those functions.

Change	Why
Loaders for qrels and run files	Part B data was typed into Python, and Part C uses TREC-format files.
Renumbering query IDs	Query numbers in the file run 1, 2, 4, 8 and so on up to 365, but the qrels number queries 1 to 225 by position. Without the fix, BM25's MAP falls from 0.2952 to 0.0090 and no error is raised. This was the change that mattered most.
Rank cutoffs	Precision on lists of up to 1000 is close to meaningless (BM25: 0.0081), so I added Precision, Recall and F1 at rank 10.
nDCG for all queries, with binary gain	Part B had grades for Q1 only, and the copy of the qrels I could get is binary.

2. Did the real-world results match your expectations?

Partly. I expected BM25 to clearly beat TF-IDF and the Language Model to come close to BM25. Neither happened.

BM25 and TF-IDF were closer than I expected. BM25's MAP is higher by only 0.0021, which is not significant (randomization $p = 0.7474$). TF-IDF even has the higher MRR, and it has the single best AP on more queries (99 against 75). Lucene's TF-IDF includes length normalization, which may remove much of the gap a basic TF-IDF would show, though I did not test that.

The Language Model did worse than I expected. Its MAP is 0.0611 below BM25's ($p < 10^{-5}$). The default μ=2000 is about 20 times the mean document length of 102 terms. A smaller μ helped (MAP 0.2757 at μ=50), but it still stayed below BM25.

The absolute scores did look reasonable for a small, hard collection. Queries average only 7.2 relevant documents, so P@10 cannot be high on many of them.

3. Which metrics were most informative for your experiment, and why?

Metric	What it told me
MAP	The most informative. It uses the position of every relevant document and clearly separated the Language Model.
nDCG@10	Same conclusion as MAP, focused on the first page. It told me less than usual because the gains were binary.
MRR	A different view: TF-IDF and BM25 swap places on it, although that difference is not significant.
P@10, Recall@10	Easy to read, but limited by how many relevant documents a query has (from 1 to 39).
Full-list P, R, F1	The least informative. They are almost identical across models (Recall 0.951 to 0.952) because each list holds nearly every document that matches any query term.

The paired significance tests were as useful as the metrics. Without them I could have read the small BM25 and TF-IDF gap as a real difference.

4. What are the main limitations of your evaluation?

1. The collection is old and narrow. It has 1960s aeronautics abstracts and long questions (about 18 words), which do not represent modern web or enterprise search.
2. Relevance is binary in the copy I used, so nDCG did not measure graded relevance. Rows labeled "no interest" in the original were treated as non-relevant.
3. Unjudged documents count as non-relevant, which can understate any model.
4. The main comparison uses default parameters. I picked the sensitivity settings after seeing test queries and had no separate tuning set, so those scores are not clean estimates.
5. Results depend on Lucene's implementations and on `EnglishAnalyzer`, whose small stop list keeps words such as "what".
6. These offline metrics say nothing about speed, satisfaction or real click behavior.